



Transparency & Accountability: from bias mitigation to trustworthy systems

Ethics & bias in AI: week 8

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Overview

- Assignment – notes and guidance

Lecture Transparency & Accountability

- Introduction to transparency and accountability
 - Regulatory aspects relating to transparency
- Case study: why transparency matters
- Data documentation
 - Case study: Models cards
 - How widely adopted?
- Explainability for AI
- Limitations and critiques

Key questions

- What are the potential consequences of unaccountable and opaque AI?
- How can we ensure AI systems align with human values and societal norms?
- Is simply creating mechanisms for transparency enough?
- What are the ongoing challenges and obstacles for transparency initiatives?



The Regulatory & Ethical Landscape - Demanding Accountability

GDPR (General Data Protection Regulation): Emphasises the 'right to explanation' in automated decision-making, requiring transparency and accountability.

AI Ethics Guidelines: Developed by various organizations (e.g., EU, OECD), these guidelines promote principles like fairness, transparency, and human oversight.

Proposed AI Acts: Many countries are working on specific legislation to regulate high-risk AI applications.

Transparency in the EU AI Act

“transparency means that AI systems are developed and used in a way that **allows appropriate traceability and explainability**, while **making humans aware** that they communicate or interact with an AI system, as well as **duly informing deployers** of the **capabilities** and **limitations** of that AI system **and affected persons** about their **rights**” (Recital 27).

Source: <https://www.euaiact.com/key-issue/5>

EU AI ACT Transparency

General purpose AI systems

1. Creating technical documentation for GPAI models, covering their training, testing, and evaluation processes
2. Supplying information and documentation to AI system providers who seek to use the GPAI model in their products, helping them understand the model's capabilities and limitations to meet their legal obligations
3. Providing a detailed summary of the training content and data to enhance transparency

High Risk AI systems

1. design and develop their HRAIS in a way that ensures sufficient transparency for deployers to reasonably understand the system's functioning and output accordingly
2. provide 'instructions for use' to deployers along with HRAIS, which must give clear and complete information on the characteristics, functioning and other key features of HRAIS to deployers

EU AI ACT Transparency

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The Type of AI System	Obligation	Exception
AI Systems in Direct Interaction with Human Beings	The provider must inform individuals about their interaction with an AI system if it is not readily apparent to the user.	AI systems authorized by law for criminal offense prevention, detection, investigation, or prosecution (collectively “law enforcement”), provided they are not publicly available for reporting criminal offenses
AI Systems Generating Synthetic Content	The provider must ensure that synthetic AI outputs are labeled in a machine-readable format and identifiable as artificially generated or manipulated.	- AI systems used for standard editing assistance - AI systems that do not significantly alter the input data or its meaning - AI systems authorized for law enforcement purposes
AI Systems for Emotion Recognition or Biometric Categorization	The deployer must: - Inform individuals exposed to the system about its operation - process personal data in compliance with the relevant data protection laws (e.g., GDPR)	AI systems authorized for law enforcement purposes
AI Systems Generating Deep Fake	The deployer must disclose to individuals that the content has been artificially produced or modified.	- Artistic or creative works where transparency does not hinder enjoyment - AI systems authorized for law enforcement purposes
AI Systems Generating/ Manipulating Text of Public Interest	The deployer must disclose that the text, which is published for the purpose of informing the public on matters of public interest, is artificially created or modified.	- AI-generated content that has been subject to human review or editorial control, with the editorial responsibility lying with a natural or legal person - AI systems authorized for law enforcement purposes

Source: <https://www.euaiact.com/key-issue/5>

Why transparency matters

Case study of Stable Diffusion generating harmful content due to issues in the training data

LAION-5B image dataset

- An open dataset created by the Large-scale Artificial Intelligence Open Network that's popular among researchers and used in some generative AI systems.
- LAION-5B dataset represents nearly 6 billion samples, which include URLs, descriptions, and other data that might be associated with an image scraped from the internet.
- Publicly accessible generative AI models, like Stable Diffusion XL1.5 and MidJourney have been trained on datasets containing CSAM such as LAION-5B

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RELEASING RE-LAION 5B: TRANSPARENT ITERATION ON LAION-5B WITH ADDITIONAL SAFETY FIXES

by: LAION e.V., 30 Aug, 2024

Today, following [a safety revision procedure](#), we announce Re-LAION-5B, an updated version of LAION-5B, that is the first web-scale, text-link to images pair dataset to be thoroughly cleaned of known links to suspected CSAM.

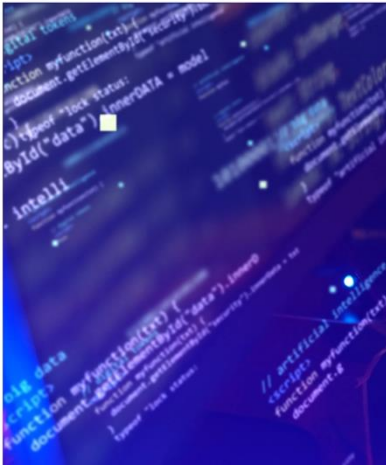
<https://laion.ai/blog/relaion-5b/>

You can read more on this case study from the [Internet Watch Foundation report](#)

Child sexual abuse material found on popular dataset shows risks for federal AI research

The National Science Foundation doesn’t know whether or if its researchers have used a dataset containing child sexual abuse material, which was flagged in a recent Stanford report.

BY REBECCA HEILWEIL • JANUARY 22, 2024



Stanford

Cyber Policy Center
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All Cyber News / Blogs / December 20, 2023

Investigation Finds AI Image Generation Models Trained on Child Abuse

A new report identifies hundreds of instances of exploitative images of children in a public dataset used for AI text-to-image generation models.

David Thiel

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in

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<https://fedscoop.com/ai-federal-research-database-laion-csam/>
<https://cyber.fsi.stanford.edu/news/investigation-finds-ai-image-generation-models-trained-child-abuse>

LAION-5B image dataset

- Individuals, organisations , researchers etc., who downloaded this dataset may have unwittingly downloaded illegal materials and perpetuated harms.
- Industry and academia must give more substantial consideration to the datasets accepted for usage in the field.
- LAION-5B was submitted to the highly influential Neural Information Processing Systems (NeurIPS) conference for peer review,
 - it was accepted and awarded the best paper in the “datasets and benchmarks” track.

Can you think of other examples we discussed that might have had different outcomes if better documentation and error reporting had been required?

Initiatives and methods to
advance transparency

Fairness toolkits

- Toolkits aim to
 - make the identification and mitigation of bias in AI systems and datasets more **accessible**
 - to advance towards more widespread adoption of ethical practices.
- A number of toolkits have been released. These typically:
 - aim to serve different audiences.
 - implement different measures of fairness.

Examples of fairness toolkits

- [Fairkit-learn](#) - Fairkit-learn is an open-source, publicly available Python toolkit designed to help data scientists evaluate and explore machine learning models with respect to quality and fairness metrics simultaneously. Built on top of scikit-learn.
- [Aequitas' Bias & Fairness Audit Toolkit](#) - The Bias Report is powered by Aequitas, an open-source bias audit toolkit for machine learning developers, analysts, and policymakers to audit machine learning models for discrimination and bias, and make informed and equitable decisions around developing and deploying predictive risk-assessment tools.
- [IBM's AI Fairness 360 Open Source Toolkit](#) - This is IBM's open-source toolkit that includes approach to report, and mitigate discrimination and bias in machine learning models.
- [Microsoft Fairlearn](#) - An open-source toolkit for assessing and improving fairness in machine learning products developed by Microsoft – associated [paper](#)
- [What-if tool](#) – A toolkit developed by Google research for investigating model performance and fairness. You can also explore counterfactual examples (i.e. for classification models, compare any example to the example most-similar to it but which is classified as a different).

Industry-led responsible AI tools

Company	Tool	Description	Type	Principles
Microsoft	Human AI Interaction (HAX) Toolkit	HAX design guidelines, patterns, and failure detection tool	Guideline, software tool	Explainability, contestability
	AI Trust Score	Understanding AI feature	Tool	Explainability
	Fairness checklist	A fairness checklist	Guideline	Fairness
	Fairlearn	Assess and mitigate fairness issues	Software tool	Fairness
	InterpretML	Package used for explaining models	Software tool	Explainability
	Counterfeit	Assess AI security risks	Software tool	Security
	Conversational AI guidelines	Guidelines for developing responsible conversational AI	Guideline	Explainability, privacy
	SmartNoise	Differential privacy	Software tool	Privacy
	Presidio	Library for data privacy protection	Software tool	Privacy
	Datasheet for Datasets	Documents datasets	Tool	Transparency, explainability, accountability
	Confidential computing for ML	Provides trusted execution environments or encryption	Software tool	Privacy, security
	SEAL	Homomorphic encryption library	Software tool	Privacy
	Responsible AI toolkit	Dashboards for model explainability	Software tool	Explainability
Google	People + AI Guidebook (PAIR)	23 design patterns	Guideline	Explainability, privacy, reliability
	Rules of Machine Learning	Best Practices for developing models	Guideline	Reliability
	Human-Centered Machine Learning	Steps to design models	Guideline	Explainability, reliability
	Model cards	Reporting on model meta data	Tool	Transparency, explainability
	Data cards	Creating dataset reports	Tool	Transparency, explainability
	Fairness indicators	Evaluate fairness metrics	Software tool	Fairness
	Know your data	Data visualisation tool for fairness	Software tool	Fairness
	ML-fairness-gym	Exploring long-term fairness impacts	Software tool	Fairness
	Language Interpretability Tool	Visualization of NLP models	Software tool	Explainability
	What-If tool	Visualize model behavior	Software tool	Explainability, fairness
	Explainable AI	Interpret predictions	Software tool	Explainability
	Google Tensorflow Privacy	Providing differential privacy	Software tool	Privacy
	Google TensorFlow Federated	Privacy-preserving machine learning	Software tool	Privacy

IBM	AI Explainability 360	Explaining machine learning models	Software tool	Explainability
	AI Fairness 360	Bias-mitigation tool	Software tool	Fairness
	AI Privacy 360	Tools for privacy protection	Software tool	Privacy
	Adversarial Robustness 360	Overcoming adversarial attacks	Software tool	Security
	AI FactSheets 360	To make AI models transparent	Tool	Transparency, explainability
	Uncertainty Quantification 360	Tools to test how reliable AI predictions	Software tool	Reliability
	Causal Inference 360	Causal inference analysis	Software tool	Explainability
	Fairness Flow	A technical toolkit for fairness	Software tool	Fairness
Meta	AI System Cards	Explains AI systems	Tool	Transparency, explainability
	Crypten	Securing & privacy-preserving models	Software tool	Privacy, security
	Captum	A model interpretability library	Software tool	Explainability
Amazon	Amazon SageMaker Clarify	Visualising training data and models	Software tool	Explainability, fairness
	Amazon SageMaker Model Monitor	Detecting inaccurate predictions	Software tool	Reliability
	Amazon Augmented AI	Human review of predictions	Software tool	Reliability, contestability

Source: <https://dl.acm.org/doi/pdf/10.1145/3626234>

Lu, Q., Zhu, L., Xu, X., Whittle, J., Zowghi, D., & Jacquet, A. (2023). Responsible ai pattern catalogue: A collection of best practices for ai governance and engineering. ACM Computing Surveys.

“☑ Fairness Toolkits, A Checkbox Culture?”

- <https://dl.acm.org/doi/10.1145/3600211.3604674>

The screenshot shows the ACM Digital Library interface. At the top, there are logos for ACM Digital Library and the Association for Computing Machinery, along with navigation links for Browse, About, and Sign in. Below this is a horizontal menu with links for Journals, Magazines, Proceedings, Books, SIGs, Conferences, and People. A search bar labeled 'Search ACM Digital Library' is positioned on the right. Underneath the menu is a secondary navigation bar with links for Conference, Proceedings, Upcoming Events, Authors, Affiliations, and Award Winners. The main content area displays the breadcrumb trail: Home > Conferences > AIES > Proceedings > AIES '23 > "☑ Fairness Toolkits, A Checkbox Culture?" On the Factors that Fragment Developer Practices in Handling Algorithmic Harms. The article is identified as a RESEARCH-ARTICLE, OPEN ACCESS, and a trophy icon is present. Social media sharing icons for X, LinkedIn, GitHub, Facebook, and Email are shown. The title of the paper is prominently displayed. Below the title, the authors are listed: Agathe Balayn, Mireia Yurrita, Jie Yang, and Ujwal Gadiraju, with a link to 'Authors Info & Claims'. The publication details state it is from 'AIES '23: Proceedings of the 2023 AAAI/ACM Conference on AI, Ethics, and Society', pages 482 - 495, with the DOI link <https://doi.org/10.1145/3600211.3604674>. The publication date is 29 August 2023, and there is a link to the 'Publication History'. A 'Check for updates' button is also visible. At the bottom, there are citation statistics showing 11 citations and 1,069 views, along with buttons for 'All formats' and 'PDF'.

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"☑ Fairness Toolkits, A Checkbox Culture?" On the Factors that Fragment Developer Practices in Handling Algorithmic Harms

Authors: Agathe Balayn, Mireia Yurrita, Jie Yang, Ujwal Gadiraju | [Authors Info & Claims](#)

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Published: 29 August 2023 [Publication History](#)

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Documentation

The Data Cards Playbook

USER GUIDE

ACTIVITIES

PATTERNS

FOUNDATIONS

The Data Cards Playbook



A toolkit for transparency in AI dataset
documentation.

Documentation – benefits and barriers

Discuss with you neighbour. Consider the following:

- What are the benefits of creating documentation for datasets and models? How might they be useful?
- Can you think of any projects or contexts where the creation of a datasheet might not be permitted?
- Can you foresee any practical challenges or potential objections to integrating datasheets into the development process?

Documentation practices

Different approaches to documentation have been offered, e.g.

- [Data Statements](#) (Bender and Friedman, 2018)
- [Datasheets for Datasets](#) (Gebru et al. 2018; Gebru et al. 2020)
- [Dataset Nutrition Labels](#) (Holland et al. 2018; Chmielinski et al. 2020)
- [FactSheets](#) (Arnold et al. 2019)
- [Google's Data Cards playbook](#)
- [Data Cards](#) (Pushkarna et al. 2022)
- [Model Cards for Model Reporting](#) (Mitchell et al. 2019)
- [Nutrition Labels for Data and Models](#) (Stoyanovich and Howe 2019)

Model- and system- level transparency

- Model Cards (Mitchell et al. 2019; OECD 2022)
- System Cards (Meta 2022)
- UK's Algorithmic Transparency Reporting Standard (CDDO and DSIT 2024)
- Often advised that developers incorporate data-level transparency approaches (e.g. data cards) into their model- and system- level transparency approaches
 - Although not fully sure how often this is completed.

Model Card - Smiling Detection in Images

Model Details

- Developed by researchers at Google and the University of Toronto, 2018, v1.
- Convolutional Neural Net.
- Pretrained for face recognition then fine-tuned with cross-entropy loss for binary smiling classification.

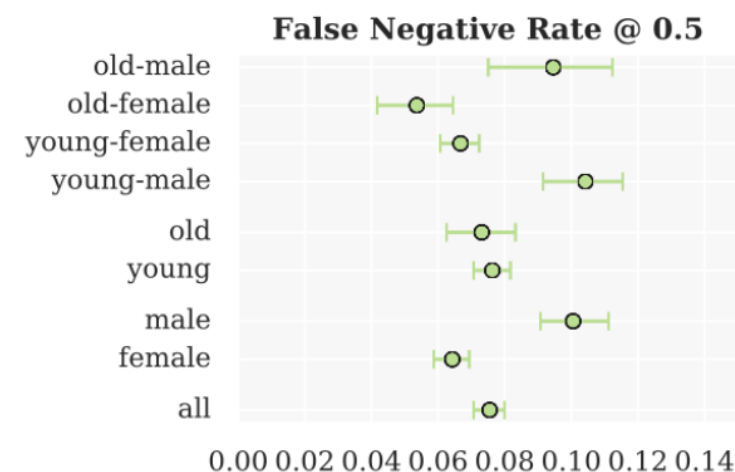
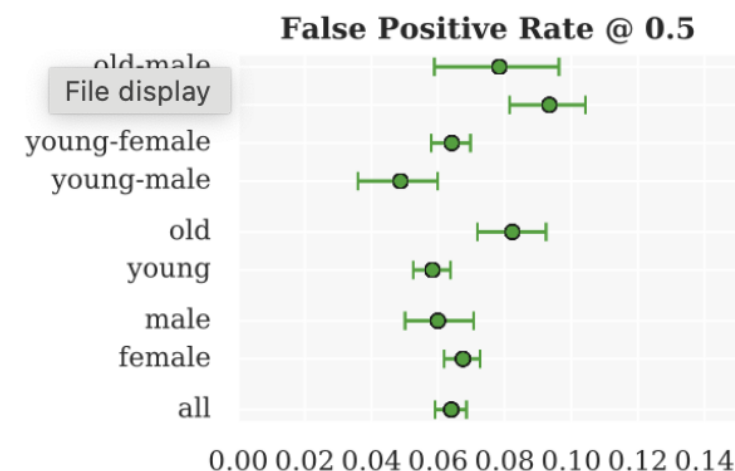
Intended Use

- Intended to be used for fun applications, such as creating cartoon smiles on real images; augmentative applications, such as providing details for people who are blind; or assisting applications such as automatically finding smiling photos.
- Particularly intended for younger audiences.
- Not suitable for emotion detection or determining affect; smiles were annotated based on physical appearance, and not underlying emotions.

Factors

- Based on known problems with computer vision face technology, potential relevant factors include groups for gender, age, race, and Fitzpatrick skin type; hardware factors of camera type and lens type; and environmental factors of lighting and humidity.
- Evaluation factors are gender and age group, as annotated in the publicly available dataset CelebA [36]. Further possible factors not currently available in a public smiling dataset. Gender and age determined by third-party annotators based on visual presentation, following a set of examples of male/female gender and young/old age. Further details available in [36].

Quantitative Analyses



False Discovery Rate @ 0.5

Metrics

- Evaluation metrics include **False Positive Rate** and **False Negative Rate** to measure disproportionate model performance errors across subgroups. **False Discovery Rate** and **False Omission Rate**, which measure the fraction of negative (not smiling) and positive (smiling) predictions that are incorrectly predicted to be positive and negative, respectively, are also reported. [48]
- Together, these four metrics provide values for different errors that can be calculated from the confusion matrix for binary classification systems.
- These also correspond to metrics in recent definitions of “fairness” in machine learning (cf. [6, 26]), where parity across subgroups for different metrics correspond to different fairness criteria.
- 95% confidence intervals calculated with bootstrap resampling.
- All metrics reported at the .5 decision threshold, where all error types (FPR, FNR, FDR, FOR) are within the same range (0.04 - 0.14).

Training Data

- CelebA [36], training data split.

Evaluation Data

- CelebA [36], test data split.
- Chosen as a basic proof-of-concept.

Ethical Considerations

- Faces and annotations based on public figures (celebrities). No new information is inferred or annotated.

Caveats and Recommendations

- Does not capture race or skin type, which has been reported as a source of disproportionate errors [5].
- Given gender classes are binary (male/not male), which we include as male/female. Further work needed to evaluate across a spectrum of genders.
- An ideal evaluation dataset would additionally include annotations for Fitzpatrick skin type, camera details, and environment (lighting/humidity) details.

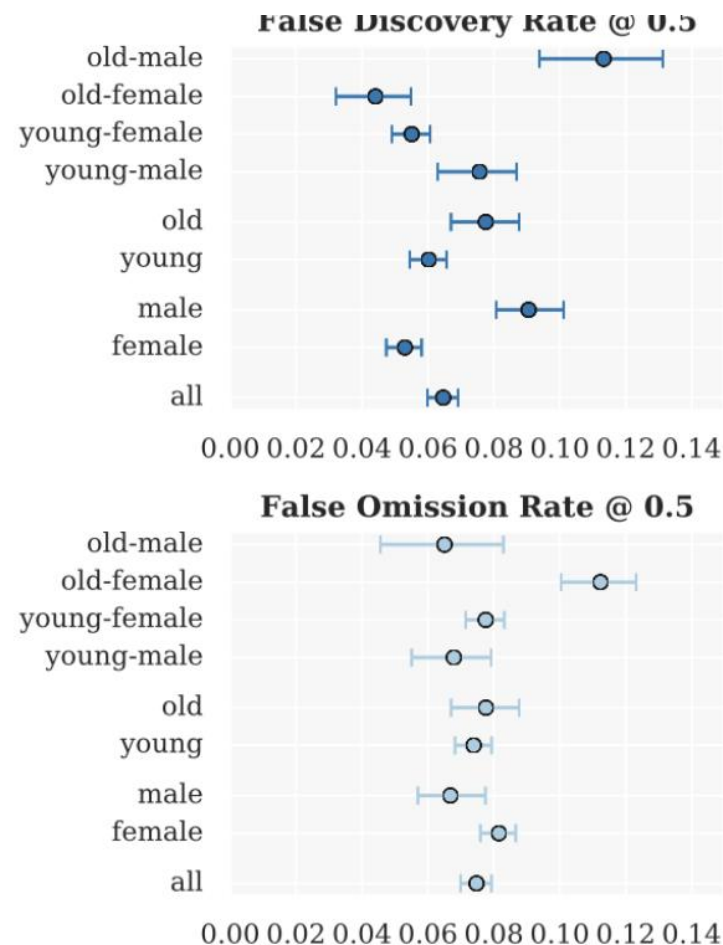


Figure 2: Example Model Card for a smile detector trained and evaluated on the CelebA dataset.

- <https://dl.acm.org/doi/10.1145/3285573>

Model Card - Toxicity in Text

Model Details

- The TOXICITY classifier provided by Perspective API [32], trained to predict the likelihood that a comment will be perceived as toxic.
- Convolutional Neural Network.
- Developed by Jigsaw in 2017.

Intended Use

- Intended to be used for a wide range of use cases such as supporting human moderation and providing feedback to comment authors.
- Not intended for fully automated moderation.
- Not intended to make judgments about specific individuals.

Factors

- Identity terms referencing frequently attacked groups, focusing on sexual orientation, gender identity, and race.

Metrics

- Pinned AUC, as presented in [11], which measures threshold-agnostic separability of toxic and non-toxic comments for each group, within the context of a background distribution of other groups.

Ethical Considerations

- Following [31], the Perspective API uses a set of values to guide their work. These values are Community, Transparency, Inclusivity, Privacy, and Topic-neutrality. Because of privacy considerations, the model does not take into account user history when making judgments about toxicity.

Training Data

- Proprietary from Perspective API. Following details in [11] and [32], this includes comments from a online forums such as Wikipedia and New York Times, with crowdsourced labels of whether the comment is “toxic”.
- “Toxic” is defined as “a rude, disrespectful, or unreasonable comment that is likely to make you leave a discussion.”

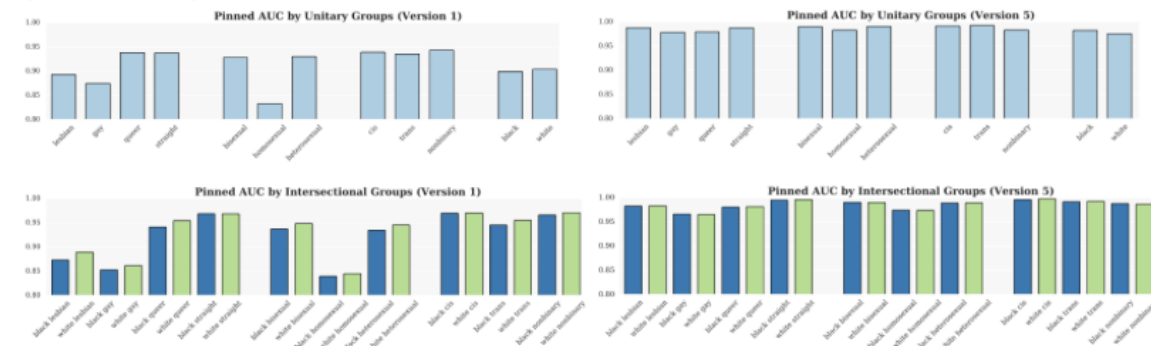
Evaluation Data

- A synthetic test set generated using a template-based approach, as suggested in [11], where identity terms are swapped into a variety of template sentences.
- Synthetic data is valuable here because [11] shows that real data often has disproportionate amounts of toxicity directed at specific groups. Synthetic data ensures that we evaluate on data that represents both toxic and non-toxic statements referencing a variety of groups.

Caveats and Recommendations

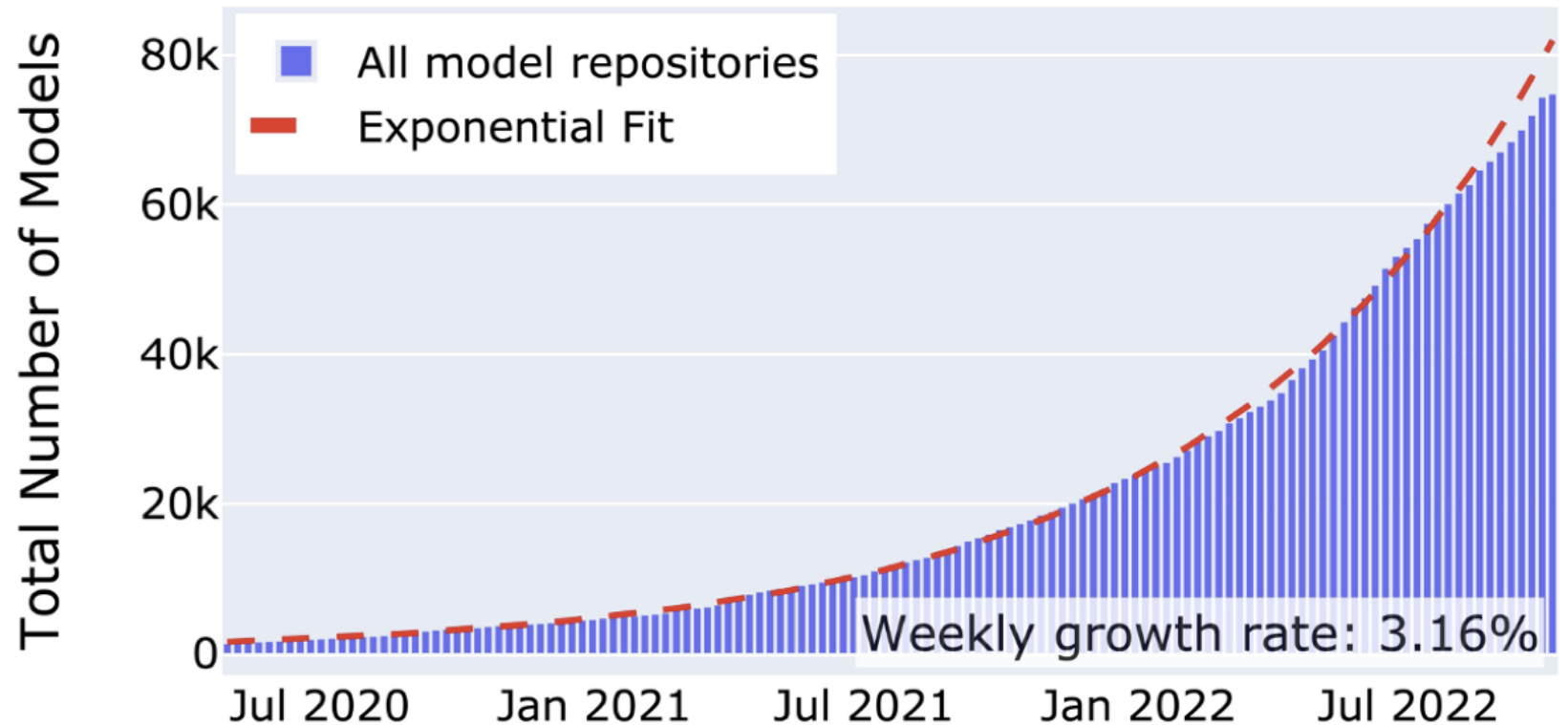
- Synthetic test data covers only a small set of very specific comments. While these are designed to be representative of common use cases and concerns, it is not comprehensive.

Quantitative Analyses



The rise of pre-trained models

- Low barrier to access and implement models
- Exponential growth of models available via platforms such as hugging face



How widely are Model cards used on hugging face?

What's documented in AI? Systematic Analysis of 32K AI Model Cards (Liang et al 2024)

- Analysed 74,970 AI model repositories on Hugging Face uploaded by 20,455
- What percentage do you think had accompanying model cards?

<https://arxiv.org/pdf/2402.05160>



[Poll everywhere:](https://PollEv.com/multiple_choice_polls/CIQFEJlKbYxfVfy1uHUeE/respond)

https://PollEv.com/multiple_choice_polls/CIQFEJlKbYxfVfy1uHUeE/respond

Poll: what percentage of models have model cards?



[<](#) Activities

 Visual settings

 Edit

When poll is active respond at

PollEv.com
/shaunaconcannon930

Send **shaunaconcannon930** to
07480 781235



What percentage of models on hugging face examined had accompanying model cards?

Powered by  **Poll Everywhere**

Rank the sections of a model card by ethical importance

- https://www.polleverywhere.com/ranking_polls/bjhOnvk4fG3zBP5a_mkRPi



Poll: Rank the model card sections in order of (ethical) importance



[< Activities](#)

 Visual settings

 Edit



When poll is active respond at

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Rank the model card sections in order of (ethical) importance

 0

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Open challenges to transparency

AI Data Transparency: an Exploration Through the Lens of AI Incidents (Worth et al 2024)

- Evidence suggests that use of these documentation practices often remains ad-hoc and inconsistent (Yang, Liang, and Zou 2024; Liang et al. 2022).
- One Microsoft study demonstrated that UX designers experience a lack of data transparency information to understand pre-trained models for responsibly implementing these in AI products (Liao et al. 2023)
- The Stanford Foundation Model Transparency Index demonstrated that the data transparency of a selection of 10 popular foundation models is very low (Bommasani et al. 2023b).
- More broadly, there remains limited further evidence on how far the range of data transparency approaches have permeated to help different stakeholders understand AI systems and the role of data in their downstream impacts.

AI Data Transparency: an Exploration Through the Lens of AI Incidents (Worth et al 2024)

- Used the AI incidents database (AIID) generated by the Partnership on AI to identify AI systems associated with public concern.
- Sample of 135 incidents between January 2022 - March 2024, which generated a sample of 54 AI systems
- Explored the data transparency of AI systems underlying the sample of incidents.
- Found low levels of AI data transparency across 25 models
- Identified issues of transparency and explainability at model- and system levels that create barriers for investigating data transparency information.

Liesenfeld, Lopez, Dingemanse (2023)

- Review the risks of relying on proprietary software
- Review best practices for open, transparent and accountable 'AI'
- Find over 20 ChatGPT alternatives at varying degrees of openness, development and documentation
- Tech is never a *fait accompli* unless we make it so
- Openness enables critical computational literacy

Liesenfeld, Andreas, Alianda Lopez, and Mark Dingemanse. 2023. "Opening up ChatGPT: Tracking Openness, Transparency, and Accountability in Instruction-Tuned Text Generators." In CUI '23: Proceedings of the 5th International Conference on Conversational User Interfaces. July 19-21, Eindhoven.

Project (maker, bases, URL)	Availability						Documentation				Access			
	Open code	LLM data	LLM weights	RL data	RL weights	License	Code	Architecture	Preprint	Paper	Modelcard	Datasheet	Package	API
WizardLM-7B Microsoft & Peking Univer...	~	~	✗	✓	~	~	~	✓	✓	✗	✗	✗	✗	✗
	LLM base: LLaMA-7B		RL base: Evol-Instruct (synthetic)											\$
StableVicuna-13B CarperAI	~	✗	~	~	~	~	~	~	~	✗	~	✗	✗	~
	LLM base: LLaMA		RL base: OASST1 (human), GPT4All (h...											\$
Falcon-40B-instruct Technology Innovation In...	✗	~	✓	~	✗	✓	✗	~	~	✗	~	✗	✗	✗
	LLM base: Falcon 40B		RL base: Baize (synthetic)											\$
UltraLM OpenBMB	✗	✗	~	✓	~	✗	✗	~	✓	✗	~	~	✗	✗
	LLM base: LLaMA2		RL base: UltraFeedback (part synthetic)											\$
Koala 13B BAIR	✓	~	~	~	✗	~	~	~	✗	✗	✗	✗	✗	✗
	LLM base: LLaMA 13B		RL base: HC3, ShareGPT, alpaca (synt...											\$
Stable Beluga 2 Stability AI	✗	✗	~	✗	✓	~	✗	~	~	✗	~	✗	✗	~
	LLM base: LLaMA2		RL base: Orca-style (synthetic)											\$
Stanford Alpaca Stanford University CRFM	✓	✗	~	~	~	✗	~	✓	✗	✗	✗	✗	✗	✗
	LLM base: LLaMA		RL base: Self-Instruct (synthetic)											\$
Falcon-180B-chat Technology Innovation In...	✗	~	~	~	~	✗	✗	~	~	✗	~	✗	✗	✗
	LLM base: Falcon 180B		RL base: OpenPlatypus, Ultrachat, Airo...											\$
LLaMA2 Chat Facebook Research	✗	✗	~	✗	~	✗	✗	~	~	✗	~	✗	✗	~
	LLM base: LLaMA2		RL base: Meta, StackExchange, Anthropic											\$
Solar 70B Upstage AI	✗	✗	~	✗	~	✗	✗	✗	✗	✗	~	✗	✗	~
	LLM base: LLaMA2		RL base: Orca-style, Alpaca-style											\$
ChatGPT OpenAI	✗	✗	✗	✗	✗	✗	✗	✗	~	✗	~	✗	✗	✗
	LLM base: GPT 3.5		RL base: Instruct-GPT											\$
Xwin-LM Xwin-LM	✗	✗	~	✗	✗	✗	✗	✗	✗	✗	✗	✗	✗	~
	LLM base: LLaMA2		RL base: unknown											\$

Liesenfeld, Andreas, Alianda Lopez, and Mark Dingemanse. 2023. "Opening up ChatGPT: Tracking Openness, Transparency, and Accountability in Instruction-Tuned Text Generators." In CUI '23: Proceedings of the 5th International Conference on Conversational User Interfaces. July 19-21, Eindhoven.

Project (maker, bases, URL)	Availability						Documentation						Access	
	Open code	LLM data	LLM weights	RL data	RL weights	License	Code	Architecture	Preprint	Paper	Modelcard	Datasheet	Package	API
BLOOMZ bigscience-workshop	✓	✓	✓	✓	~	~	✓	✓	✓	✓	✓	✓	✗	✓
	LLM base: BLOOMZ, mT0			RL base: xP3										\$
Open Assistant LAION-AI	✓	✓	✓	✓	✗	✓	✓	✓	~	✗	✗	✗	✓	✓
	LLM base: Pythia 12B			RL base: OpenAssistant Conversations										\$
Pythia-Chat-Base-7... togethercomputer	✓	✓	✓	✓	✗	✓	✓	✓	~	✗	~	~	✓	✗
	LLM base: EleutherAI pythia			RL base: OIG										\$
RedPajama-INCITE... TogetherComputer	~	✓	✓	✓	✓	~	~	~	✗	✗	✓	✓	✗	~
	LLM base: RedPajama-INCITE-7B-Base RL base: various (GPT-JT recipe)													\$
dolly databricks	✓	✓	✓	✓	✗	✓	✓	✓	~	✗	✗	✗	✓	✗
	LLM base: EleutherAI pythia			RL base: databricks-dolly-15k										\$
OpenChat 3.5 7B Tshinghua University	✓	✗	✓	✗	✓	✓	~	✓	✓	✗	~	✗	✓	~
	LLM base: Mistral 7B			RL base: ShareGPT with C-RLFT										\$
MPT-30B Instruct MosaicML	✓	~	✓	~	✗	✓	✓	~	✗	✗	~	✗	✓	~
	LLM base: MosaicML			RL base: dolly, anthropic										\$
MPT-7B Instruct MosaicML	✓	~	✓	~	✗	✓	✓	~	✗	✗	✓	✗	✓	✗
	LLM base: MosaicML			RL base: dolly, anthropic										\$
trlx carperai	✓	✓	✓	~	✗	✓	✓	~	✗	✗	✗	✗	~	✓
	LLM base: various (pythia, flan, OPT)			RL base: various										\$
Vicuna 13B v 1.3 LMSYS	✓	~	✓	✗	✗	~	✓	✗	✓	✗	~	✗	✓	~
	LLM base: LLaMA			RL base: ShareGPT										\$
minChatGPT ethanyanjiali	✓	✓	✓	~	✗	✓	✓	~	✗	✗	✗	✗	✗	✓
	LLM base: GPT2			RL base: anthropic										\$

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Liesenfeld, Lopez, Dingemanse (2023)

- Argue: openness is key
 - for fundamental research,
 - for fostering critical computational literacy,
 - for making informed choices for or against deployment of instruction-tuned LLM architectures.
- Open alternatives provide ways to build reproducible workflows, chart resource costs, and lessen reliance on corporate whims.

Liesenfeld, Lopez, Dingemanse (2023)

- Openness is *not* the full solution to the scientific and ethical challenges of conversational text generators.

“Open data will not mitigate the harmful consequences of thoughtless deployment of large language models, nor the questionable copyright implications of scraping all publicly available data from the internet.”

Summary: open challenges

- Transparency is increasingly recognized as important, but adoption of transparency practices is uneven. Some factors:
 - Inherent opaqueness of AI systems makes explanations challenging (more on this next week)
 - Proprietary vs open datasets and models motivate different levels of transparency
 - Efforts to create transparent documentation may not be part of existing workflows or practices
 - Many different approaches to transparency – which to choose?
 - Available does not equal accessible - if documentation is available, is it interpretable and by whom? What s the quality of explanations and documentation? – currently inconsistent.